CONTROL REINFORCEMENT LEARNING WITH SAES

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WHAT IS REINFORCEMENT LEARNING?

- In reinforcement learning we are trying to learn a policy
- Often assume we enter an environment without domain knowledge
- The policy is the guide for the agent on how it should behave in a given state in an environment
- Environment is model of the world
- Trying to maximize reward received from environment

COMMON TERMINOLOGY

- Environment
- State
  - s is a state, often the current state
  - s' is the next state in your sequence
  - S is the set of all states
- Actions
  - a, a', A
- Reward
  - r is reward
  - $R(s)$ is the reward function, returning an amount of reward given upon entering state s . Sometimes noted as just R

COMMON TERMINOLOGY

- Markov Decision Process (MDP)
  - $< S, A, T, \gamma, R >$
  - T is a transition function : $T(s, a, s') \rightarrow [0,1], or\ P(s' \mid s, a)$
  - γ is a discount factor for reward
- Trying to learn a policy
  - $\pi(s)$ returns an action to taken given state s
  - Can return a probability distribution of actions
  - Can be extended to continuous action and state spaces
  - $\pi_\theta(s)$ is also common notion for deep reinforcement learning

PPO (PROXIMAL POLICY OPTIMIZATION)

- $J_\theta^{clip} = \mathbb{E}_t[\min(r_t(\theta)A_t,\ clip(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$
- $r_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta\ old}(a_t|s_t)}$ is the ratio between the new and old policy
- $A_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \ ... + (\gamma\lambda)^{T-t+1}\delta_{T-1}$
- Clip objective in a way defines a trust region
- Encourages ratio of old policy to be around $[1 - \epsilon, 1 + \epsilon]$
- When examining this function, two things control it $r_t(\theta)$ and A_t
- A_t can be positive or negative

PPO (PROXIMAL POLICY OPTIMIZATION)

- Case 1 & 2, $r_t(\theta)$ is within the clipped objective
  - $r_t(\theta)$ is returned no matter
- Case 3 & 4, $r_t(\theta)$ is lower than the clipped objective
  - If A_t is positive, $r_t(\theta)A_t$ is still returned
  - If A_t is negative, the clipped objective is returned
- Case 5 & 6, $r_t(\theta)$ is greater than the clipped objective
  - If A_t is positive, the clipped objective is returned
  - If A_t is negative, $r_t(\theta)A_t$ is still returned

INTRODUCTION

- Sparse Autoencoders (SAE) create interpretable features you can tune to adjust the behavior of a model
- Use reinforcement learning (RL) to train a model to choose which features to amplify for each token producing interpretable intervention logs
- At each step, a model takes in residual stream activations and selects a feature to amplify
- The converged policy identifies features that change model output when amplified
- Adaptive Feature Masking (AFM) used to help make sure policy doesn't converge on specific set, and tracks new features as generation occurs

REINFORCEMENT LEARNING DETAILS

- $x \in \mathbb{R}^d$ in layer ℓ where d is the hidden dimension of the layer
- The MDP is defined as < S, A, R, P >
  - $s \in S$, where $s = x \in \mathbb{R}^d$ is the residual stream activation at the layer l
  - Actions are a binary vector $a = \{0,1\}^{d_{dict}}$ where selected features are set to 1
  - P is deterministic transition function defined by the transformers forward pass
  - R is task specific reward (often +1 per correct answer, 0 else)
- $\pi_\theta(a_t \mid x_{1:t}) = \pi_\theta(a_t \mid x_t)$ where t is token generation step

REINFORCEMENT LEARNING DETAILS

- $V_\phi(s) = \mathbb{E}_{\pi_\theta}[\, r \,|s\,]$
- Implemented as MLP
- Maps $V_\phi : R^{d_{dict}} \rightarrow R^d$
- Steering co-efficient c is determined by averaging activation magnitudes for correctly predicted training samples
- PPO trained for actor and critic networks, objectives are
  - $L_{policy}(\theta) = \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon,)A_t)]$
  - $L_{critic}(\theta) = \mathbb{E}[(V_\phi(s) - r)^2]$
- $A_t = r - V_\phi(s)$

REINFORCEMENT LEARNING DETAILS

- Policy is two layer MLP with tanh activation
- Use softmax on logits produced by policy network
- During training they sample their actions from this probability, during evaluation they take the deterministic "best" option
- Use AFM to help balance exploration
- We give Boolean mask of available features to select, this way to every feature is available at all times
- $mask_i^{(t+1)} = mask_i^{(t)} \vee active_i^{(t)}, a(t) = TopK\left(\mu \odot mask^{(t)}, k\right)$
- They call their technique CRL (Control Reinforcement Learning)

EXPERIMENT

- Tested on MMLU, and applied after SFT (Supervised Fine Tuning)
- 55.2% (base) -> 55.7% (SFT) -> 56.1% (SFT + CRL)
- CRL without AFM on BBQ Ambig 62.55% (+2.39 over base) and 48.67% (+7.21 over base) on HarmBench

DISCUSSION

- For single token tasks critic network showed clear distributions between correct and incorrect samples, and BBQ had clear sample distinction
- For multi token task like GSM8K the critics evaluation climbed as the tokens were generating, meaning the model was heading in the right direction
- Divergence happened around 200 tokens in (it could tell if the answer was a good or bad one about 200 tokens in)
- Correction occurred around token 400 for poorly diverged answers (the CRL steered the answer back in the right direction)

FEATURE LEVEL INTERPRETABILITY

- Showed that the CRL activated features related to math not just on operators but other quantitative words in longer word problems
- Boosted scores across the board outside of the XSTest, which declined
- Found that the same feature for the models safety features was tied up with its reasoning about over-refusal, meaning you adjust the safety features while accommodating for over refusal
- They measured the Shannon Entropy of feature selection (can think of as how diverse the model is selecting its features) and impact rate which is fraction of how many selected features were scored as impacting the output heavily

FEATURE LEVEL INTERPRETABILITY

- Found that different tasks had different results
- Some classes preferred diverse feature selection with lower impact rates, often for longer sequenced tasks
- Some tasks used less diverse but more impact heavy feature selection
- Proves that the CRL actually explores its action space for the specific tasks rather than converge to local set of features to select

DISCUSSION

- MMLU the distribution for correct and incorrect is nearly indistinguishable , indicating a critic bottleneck
- Multi token tasks exhibit trajectory patterns throughout activation through critic value function
- Feature like “mathematical operations” activate on words like “baby” and “packs”
- Fail cases can reveal token misalignment, where tokens with no relation to math activate the math operations feature
- In successful cases, feature described as something like statistical methods activated on mathematical tokens, strengthening the models reasoning chains

DISCUSSION

- Branch point analysis is where the same problem context produces correct / incorrect outputs based on what features are chosen to activate
- Used branch point analysis to conclude that intervention depth based on your target behavior
- Early layers for syntactic / format control, and later layers for semantic reasoning
- “Feature labels provide limited predictive value for steering outcomes: “emotional relationships” solves speed calculations while “mathematical notation” fails on some arithmetic, indicating that learned steering effects diverge from semantic interpretations “

DISCUSSION

- Low H value (low feature selection diversity) might mean that the SAE does not have adequate features if it is a complex, multi step reasoning tasks. Highlights the CRL can't find enough important features to activate to adequately complete the task
- 2 axes predict CRLs effectiveness
- Signal clarity is one, CRL shows strong gains when there is room for improvement
- Second is when the desired direction is opposite of the models tendencies, it struggles here
- "Each task/layer combination requires separate PPO training (0.5–5.5 GPU-hours on a single A100, 3.3–3.8% additional parameters, 8–9% inference overhead); scaling to 100K+ dictionaries may require efficient architectures. Model evaluation is limited to Gemma-2 2B and LLaMA-3.1 8B by public SAE availability. "

CRL Steering Results

Gemma 2 2B | LLaMA 3.1 8B

TASK	LAYER	BEFORE	AFTER	GAIN
MMLU Knowledge	L24	52.06 ± 0.23	55.37 ± 0.16	+3.31
MMLU-Pro Knowledge	L25	30.30 ± 0.00	30.49 ± 0.07	+0.19
BBQ Ambig Bias	L5	60.17 ± 0.01	65.86 ± 3.03	+5.69
BBQ Disambig Bias	L5	84.38 ± 0.52	84.85 ± 0.09	+0.47
SimpleQA Knowledge	L8	3.78 ± 0.17	4.00 ± 0.29	+0.22
GSM8K Reasoning	L20	54.62 ± 0.16	55.65 ± 0.33	+1.03
HarmBench Safety	L21	41.46 ± 9.05	49.12 ± 1.59	+7.66
XSTest Safety	L12	86.35 ± 0.00	87.62 ± 0.82	+1.27

All results averaged across 3 seeds. CRL also reduces variance — e.g. HarmBench std drops from 9.05 to 1.59 on Gemma.

Token-Level Feature Activation Highlights

Corrected | Misguided

Example 1 | Example 2 | Example 3 | Example 4 | Example 5

QUESTION

Brittany and her mom go to the museum. The cost of admission is $12 for adults and $10 for children. Brittany's mom gives the cashier money for 1 child ticket and 1 adult ticket. If she received $8 in change, how much money, in dollars, did she give the cashier?

Baseline: 14 → Steered: 30 Oracle: 30

Solution:

1. **Define** variables:** Let 'x' represent the amount of money Brittany's mom gave the cashier.
2. **Set** up an equation:** The cost of the tickets is $12 (adult) + $10 (child) = $22. The change received is $8. Therefore, x - $22 = $8
3. **Solve** for x:** x = $22 + $8, x = $30

Answer: Brittany's mom gave the cashier $30

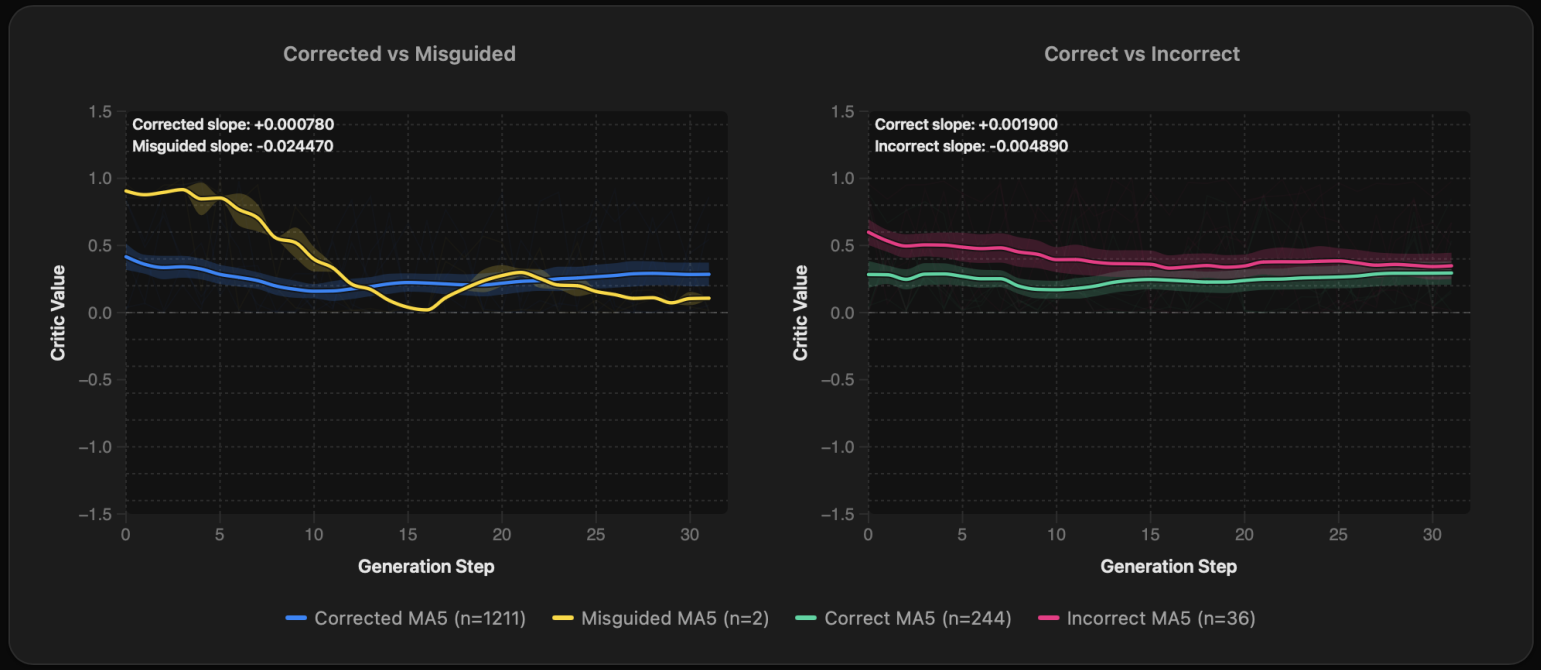
ACTIVE FEATURES

- L20/5423 numerical data & calculations 13 tok
- L20/7708 mathematical operations 10 tok
- L20/406 entities & roles 6 tok
- L20/4504 workflow & step structures 4 tok
- L20/2198 summaries & conclusions 2 tok

FEATURE ACCURACY: Steered 336/510 | Corrected 15/510 | Misguided 4/510 | Failed 174/510

Interactive Critic Trajectory Explorer

GSM8K | HarmBench | **XSTest**



Interactive Critic Trajectory Explorer

GSM8K | **HarmBench** | XSTest

Corrected vs Misguided



Correct vs Incorrect



Corrected MA5 (n=71) Misguided MA5 (n=2) Correct MA5 (n=200) Incorrect MA5 (n=88)

Interactive Critic Trajectory Explorer

GSM8K | HarmBench | XSTest

